

The 'You? Here?' effect (encounter problem)

The encounter problem is spectacular, as it provides the best evidence that the human mind is sensitive to description complexity.

Interest grows with the complexity of the place and with the simplicity of the encountered person.

Fortuitous encounters are all the more unexpected that the place is complex and the encountered person is simple.



Example: In 2008, I was travelling in Uganda. I joined a group of tourists to visit the Murchison Falls National park. I discovered that a German couple in the group were the best friends of my nephew's girlfriend. The coincidence made quite an impression on us.

Fortuitous encounters far from home

Fortuitous encounters seem to be an exception to the [rule of closeness](#), as the interest grows this time with the remoteness of the place! A proper application of the notion of unexpectedness, however, restores the prediction.

Interest grows with the *complexity of the place* and with the *simplicity of the encountered person*. The complexity of the place L is the relevant factor, not the distance: a big distant airport may be less complex than the backyard of an obscure building of a lost suburb a few kilometres away. The simplicity of the encountered person P is the relevant factor, not her closeness. Running into a celebrity may be as unexpected as running into a close colleague. These phenomena are correctly predicted by the fact that unexpectedness varies, as we will show, as:

$$U = C(L) - C(P)$$

By definition, [unexpectedness](#) U is the difference between [generation](#) complexity and [description](#) complexity: $C_w - C$.

Generation complexity C_w

Let us compute the unexpectedness of the joint presence of *ego* and P in L . Let's call $L(ego)$ and $L(P)$ the presence of self and of the encountered person P at location L . If P 's and *ego*'s common presence at L are supposed to be independent, then by definition of [independence](#):

$$C_w(L(ego) \& L(P)) = C_w(L(ego)) + C_w(L(P))$$

(note that independence is a crucial assumption for the unexpectedness of the encounter). If *ego* and P play symmetrical roles, the [W-machine](#) requires $C_w(L(ego) \& L(P)) = 2 C_w(L(ego))$ to generate the encounter.

One way to assess the term $C_w(L(ego))$ is to consider the complexity of the decisions that *ego* had to take to end up in location L . If $L(ego)$ is not itself unexpected, then in most cases, $C_w(L(ego)) = C(L)$ and amounts to the minimum size of a set of directions to reach L (exception: if L is materially difficult to reach and if L is a famous place). Finally:

$$C_w(L(ego) \& L(P)) = 2 C(L)$$

Description complexity C

The [description complexity](#) of $C(L(ego) \& L(P))$ amounts to its [conceptual complexity](#).

$$C(L(ego) \& L(P)) = C(L) + C(P)$$

and as announced, [unexpectedness](#) amounts to:

$$U(L(ego) \& L(P)) = C(L) - C(P)$$

Alternative calculus

An alternative computation of $C_w(L(P))$ consists in considering a generic method for computing $C_w(L(x))$ and then in applying it to $x=P$. Let's introduce x 's home $h(x)$. [Generating](#) $L(x)$ for x living in $h(x)$ is equivalent to generating the provenance $h(x)$ for x observed in L . For an inhabitant y of L who observes x , $h(x)$ is not unexpected: $C_w(h(x)) = C_L(h(x))$. Here, C_L means complexity as measured from L . In the case of $x=P$, we can deduce that $C_w(h(P)) = C(L)$ (supposing that $h(P)$ is as complex for y as L is for *ego*). Finally, $C_w(L(P)) = C(L)$, as previously.

This alternative calculus may be prompted by the way the anecdote is told.

- Last week I went to a small village near Bordeaux. I went to a restaurant, chose a table and asked for the menu.

Guess who was sitting at the table next to mine...

At this point, listeners may think of the identity of the encountered person based on her origin in relation to the location of the encounter.

Fortuitous encounters in your street

We supposed that L is as complex for ego as it is for P . This may not be the case. Suppose you meet a celebrity living far away in front of your door. That would be quite a coincidence! However, now, $C_w(L(ego))$ and $C(L(ego))$ are both negligible. But the value of $C_w(L(P))$ remains large. Based on the alternative calculus above, it amounts to $C(h(P))$ (the complexity of P 's home from your perspective).

Bibliography

Dessalles, J-L. (2008). [Coincidences and the encounter problem: A formal account](#). In B. C. Love, K. McRae & V. M. Sloutsky (Eds.), *Proceedings of the 30th Annual Conference of the Cognitive Science Society*, 2134-2139. Austin, TX: Cognitive Science Society.
[Note that the present account is slightly simpler than what was written at that time]

Dessalles, J-L. (2008). [La pertinence et ses origines cognitives - Nouvelles théories](#). Paris: Hermes-Science Publications.



[Back to the Simplicity Theory page.](#)